

1. A student writes several functions to swap the values of the two variables x and y, i.e. if x=1 and y=2, after calling the swap function x=2 and y=1. The different functions that the student writes are given below:

1 point

```
1 def swap1(x,y) :
2     x=y
3     y=x
4     return(x,y)
5
6 def swap2(x,y) :
7     return(y,x)
8
9 def swap3(x,y) :
10    z=x
11    x=y
12    y=z
13    return(x,y)
14
15 def swap4(x,y) :
16    x,y=y,x
17    return(x,y)
```

Which of the functions swap1, swap2, swap3, and swap4 is correct?

- ☐ None
- ☐ Function swap2 only
- ☒ Functions swap2, swap3, and swap4 only
- ☐ Functions swap3, and swap4 only

2. Consider the following two functions:

1 point

```
1 def f1(x):
2     if (x > 0):
3         x = 3*x
4         x = x / 2
5     return x
6
7 def f2(x):
8     if (x > 0):
9         x = 3*x
10    x = x / 2
11    return x
```

For what values of x will f1 and f2 return the same value?

- ☒ When x is zero or positive
- ☐ Any value of x
- ☐ For positive values of x only
- ☐ For negative values of x only

3. A recursive function in programming is a function that calls itself during its execution. The following two functions are examples of recursive functions:

1 point

```
1 def function1(length):
2     if length > 0:
3         print(length)
4         function1(length - 1)
5 def function2(length):
6     while length > 0:
7         print(length)
8         function2(length - 1)
```

What can you say about the output of function1(3) and function2(3)?

- ☐ The two programs produce the same output:
- 1
 - 2
 - 3
- ☐ function1 produces the output: 3 2 1 and function2 runs infinitely.
- ☒ function1 produces the output:
- 3
 - 2
 - 1
- and function2 runs infinitely.
- ☐ The two functions produce the same output 3 2 1.

4. The following recursive function takes three positive integer arguments:

1 point

```
1 def compute(n,x,y) :  
2     if n==0 : return x  
3     return compute(n-1,x+y,y)
```

What is the value returned by the compute function?

- ☐ $n*(x+y)$
 - ☐ x
 - ☐ $x+y$
 - ☒ $x+n*y$
5. What will the returned value be for the compute function defined in Question 4 if the argument n is negative?

1 point

- ☒ The function will never return a value.
- ☐ $x+n*y$
- ☐ $n*x+y$
- ☐ $x-n*y$

6. The following functions are all intended to check whether a string representing a dna sequence contains any characters that are not 'a','c','g','t', 'A', 'C', 'G', or 'T'. At least some of these functions are wrong. Which ones are correct?

1 point

```
1 def valid_dna1(dna):
2     for c in dna:
3         if c in 'acgtACGT':
4             return True
5         else:
6             return False
```

```
1 def valid_dna2(dna):
2     for c in dna:
3         if 'c' in 'acgtACGT':
4             return 'True'
5         else:
6             return 'False'
```

```
1 def valid_dna3(dna):
2     for c in dna:
3         flag = c in 'acgtACGT'
4     return flag
```

```
1 def valid_dna4(dna):
2     for c in dna:
3         if not c in 'acgtACGT':
4             return False
5     return True
```

- ☒ valid_dna4 only
- ☐ valid_dna2 only
- ☐ valid_dna1, valid_dna2, and valid_dna4 only
- ☐ valid_dna1, and valid_dna3 only

7. What is the type of variable L3 and what is its value if L1 and L2 are lists?

1 point

L3 = [i for i in set(L1) if i in L2]

- ☐ L3 is a set with elements common between the lists L2 and L3.
- ☒ L3 is a list that contains only the elements that are common between the lists (without duplicates).
- ☐ L3 is a tuple with elements that are both in L1 and L2
- ☐ L3 is a list with all the elements in L1 and L2

8. What will be printed after executing the following code?

1 point

```
1  >>>def f(mystring):
2      print(message)
3      print(mystring)
4      message="Inside function now!"
5      print(message)
6  >>>message="Outside function!"
7  >>>f("Test function:")
```

- ☐ Test function:
then an error message
- ☒ An error message.
- ☐ Outside function!
Test function:
Inside function now!
- ☐ Outside function!

9. Which statement below is true about a function:

1 point

- ☒ may have no parameters
- ☐ must always have a fixed number of arguments
- ☐ must always have a return statement
- ☐ must have at least one parameter

10. Which of the following function headers is correct?

1 point

- A. def afunction(a1 = 1, a2):
- B. def afunction(a1 = 1, a2, a3 = 3):
- C. def afunction(a1 = 1, a2 = 2, a3 = 3):

- ☐ None is correct.
- ☐ A,B
- ☒ C
- ☐ A

